Change of variables

(Multiple integrals)

1- Function with one variable

- Integration by parts $\leftrightarrow$ product rule.
- Chain rule for derivatives
  - $\hookrightarrow$ substitution
  - Change of variables.

Example: $f : [0;1] \to \mathbb{R}$, $x \mapsto x+1$

$$I = \int_0^1 (x+1)\,dx$$

$u \leftrightarrow x$; $u = x+1$ $\Rightarrow$ $0 \le x \le 1 \Rightarrow 1 \le u \le 2$

$\Rightarrow$ $du = dx$ $\left(\frac{du}{dx} = 1\right)$

$$\Rightarrow I = \int_1^2 u\,du$$

Chain rule: $F' = f$

$$(F \circ g)'(x) = F'(g(x)) \cdot g'(x)$$

$$\int_a^b f(x)\,dx = \int_c^d f(g(u)) \cdot g'(u)\,du \quad (*)$$

$\Rightarrow$ $x = g(u) = u - 1$

with $g : [1;2] \to [0;1]$

$u \mapsto g(u) = x$

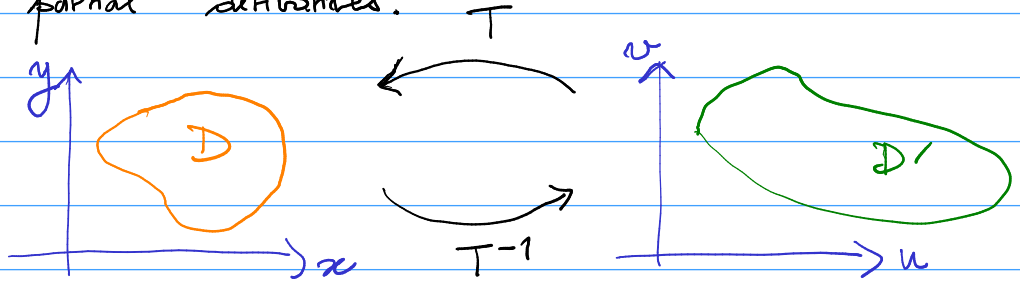
$\Rightarrow$ • $f(g(u)) = f(u-1) = u - 1 + 1 = u$

• $g'(u) = 1$

$$\Rightarrow \int_0^1 (x+1)\,dx = \int_1^2 u\,du. \quad (*)$$

2) Double integrals

For an integral $\iint_D f(x,y)\,dx\,dy$, we need a transformation $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ has continuous first partial derivatives.



$\Rightarrow$

$$T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$
$$x = g(u,v)$$
$$y = h(u,v)$$

Example:
$$\begin{cases} x = r\cos(\theta) \\ y = r\sin(\theta) \end{cases}$$
$$(r,\theta) \longmapsto (x,y)$$
$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f(r\cos(\theta), r\sin(\theta))\,\boxed{r}\,dr\,d\theta.$$

Defn: If $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ is a transformation, then the Jacobian matrix of T is given by:

$$J_T = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

The Jacobian of T is the absolute value of the determinant of J_T:

$$|J_T| = |\det(J_T)| = \left| \frac{\partial x}{\partial u}\cdot\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\cdot\frac{\partial x}{\partial v} \right|$$

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right|$$

Theorem: If $f: D \to \mathbb{R}$, $D_f \subseteq \mathbb{R}$, is continuous and $T: \mathbb{R}^2 \to \mathbb{R}^2$ is a transformation, then:

$$\iint_D f(x,y)\,dA = \iint_{D'} f(g(u,v), h(u,v))\,|J_T|\,du\,dv.$$

Example: $T: \mathbb{R}^2 \to \mathbb{R}^2$

$$T(r,\theta) = (r\cos(\theta), r\sin(\theta)) = (x,y)$$

$$J_T = \begin{pmatrix} \cos(\theta) & -r\sin(\theta) \\ \sin(\theta) & r\cos(\theta) \end{pmatrix} \quad \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix}$$

$$\Rightarrow \quad |J_T| = |r| = r$$

$$\begin{pmatrix} dx \\ dy \end{pmatrix} = J_T \cdot \begin{pmatrix} dr \\ d\theta \end{pmatrix} \iff \begin{cases} dx = \frac{\partial x}{\partial r}\,dr + \frac{\partial x}{\partial \theta}\,d\theta \\ dy = \frac{\partial y}{\partial r}\,dr + \frac{\partial x}{\partial \theta}\,d\theta \end{cases}$$

3) **Example:** • $f(x,y) = e^{\frac{x+y}{x-y}}$

$$\begin{cases} u = x+y \\ v = x-y \end{cases} \quad (T^{-1})$$

Solve for x and y

• $u+v = x+y+x-y$

$u+v = 2x$

• $u-v = x+y-(x-y)$

$= 2y$

$$\begin{cases} x = \frac{1}{2}(u+v) \\ y = \frac{1}{2}(u-v) \end{cases}$$

(1) Jacobian

$$|J_T| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$= \left|\left(\tfrac{1}{2}\right)\left(-\tfrac{1}{2}\right) - \left(\tfrac{1}{2}\right)\left(\tfrac{1}{2}\right)\right|$$

$$= \left|-\tfrac{1}{4} - \tfrac{1}{4}\right| = \left|\tfrac{2}{4}\right| = \tfrac{1}{2}$$

$$\Rightarrow \iint_D e^{\frac{x+y}{x-y}}\,dx\,dy = \iint_{D'} e^{\frac{u}{v}}\,\tfrac{1}{2}\,du\,dv.$$

(2) Assume $R = [0;1] \times [0;2]$

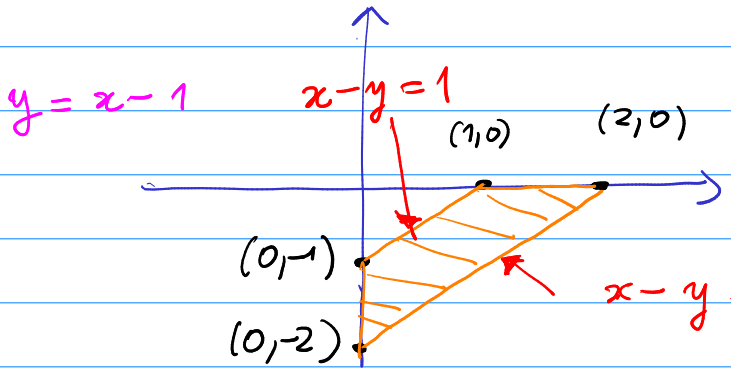
- $\left.\begin{matrix} 0 \le x \le 1 \\ 0 \le y \le 2 \end{matrix}\right\} \Rightarrow \begin{matrix} 0 \le x+y \le 3 \\ 0 \le u \le 3 \end{matrix}$

$x - y = x + (-y)$

- $\left.\begin{matrix} 0 \le x \le 1 \\ 0 \ge -y \ge -2 \end{matrix}\right\} \Rightarrow \begin{matrix} -2 \le x-y \le 1 \\ -2 \le v \le 1 \end{matrix}$

(3) D : region with vertices

$(1,0)$, $(2,0)$, $(0,-2)$, $(0,-1)$



$\Rightarrow$ $x = 0$, $y = 0$ and $x - y = 1$, $x - y = 2$

$u = x + y$
$v = x - y$

$u = -v$, $u = v$

$v = 1$ and $v = 2$

$1 \le v \le 2$

$-v \le u \le v$

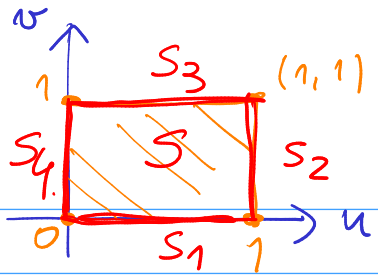
$$\iint_D e^{\frac{x+y}{x-y}}\,dx\,dy = \int_1^2 \left(\int_{-v}^{v} \frac{1}{2} e^{\frac{u}{v}}\,du \right) dv$$

- Determine the image under T:

$$\begin{cases} x = u^2 - v^2 \\ y = 2uv \end{cases}$$

of $S = \{(u,v) \mid 0 \le u \le 1, 0 \le v \le 1\}$ (rectangle)

$D \longrightarrow D'$ (new boundaries)



- $S_1 = \{(u, 0) \mid 0 \le u \le 1\}$

$$\begin{cases} x = u^2 \\ y = 0 \end{cases} \Rightarrow \underline{0 \le x \le 1}$$

- $S_2 = \{(1, v) \mid 0 \le v \le 1\}$

$$\begin{cases} x = 1 - v^2 \\ y = 2v \end{cases}$$

$0 \le v \le 1$

$0 \le 2v \le 2$

$\Rightarrow 0 \le y^2 \le 4$

$0 \le \frac{y^2}{4} \le 1$

$v = \frac{y}{2} \Rightarrow v^2 = \frac{y^2}{4} \Rightarrow$

$$\boxed{\begin{array}{c} x = 1 - \frac{y^2}{4} \\ \boxed{0 \le x \le 1} \end{array}}$$

range of $x = 1 - \frac{y^2}{4}$

- $S_3 = \{(u, 1) \mid 0 \le u \le 1\}$

$$\begin{cases} x = u^2 - 1 \\ y = 2u \end{cases}$$

(...)

$$\boxed{\begin{array}{c} x = \frac{y^2}{4} - 1 \\ \boxed{-1 \le x \le 0} \end{array}}$$

← range of x

- $S_4 = \{(0, v) \mid 0 \le v \le 1\}$

$$\begin{cases} x = -v^2 \\ y = 0 \end{cases} \Rightarrow \underline{-1 \le x \le 0} \quad (*)$$

